The Science of World-Formation: Research Programme v1.0 Core Questions, Dependency Structure, No-Go Results, Mathematical Extensions, and Experimental Roadmap

Core Questions, Dependency Structure, No-Go Results, Mathematical Extensions, and Experimental Roadmap

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Abstract

The Science of World-Formation is a research programme concerned with a prior question to ontology:

> **How can a bounded observer form, maintain, audit, and revise an operational world under incomplete representation, historical commitment, persistent purpose, and residual uncertainty?**

The programme does not begin by assuming a particular physical substrate, cosmology, symbolic tradition, or high-dimensional geometry. It begins instead from a minimal functional architecture composed of **Observer, Declaration, Purpose, Gate, Trace, Filtration, Residual, Latching, and Revision**. These components describe how a finite system selects an operationally admissible world, commits observations into history, detects mismatches between its current world and encountered evidence, preserves continuity across time, and revises either its behaviour or the declaration through which its world is represented.

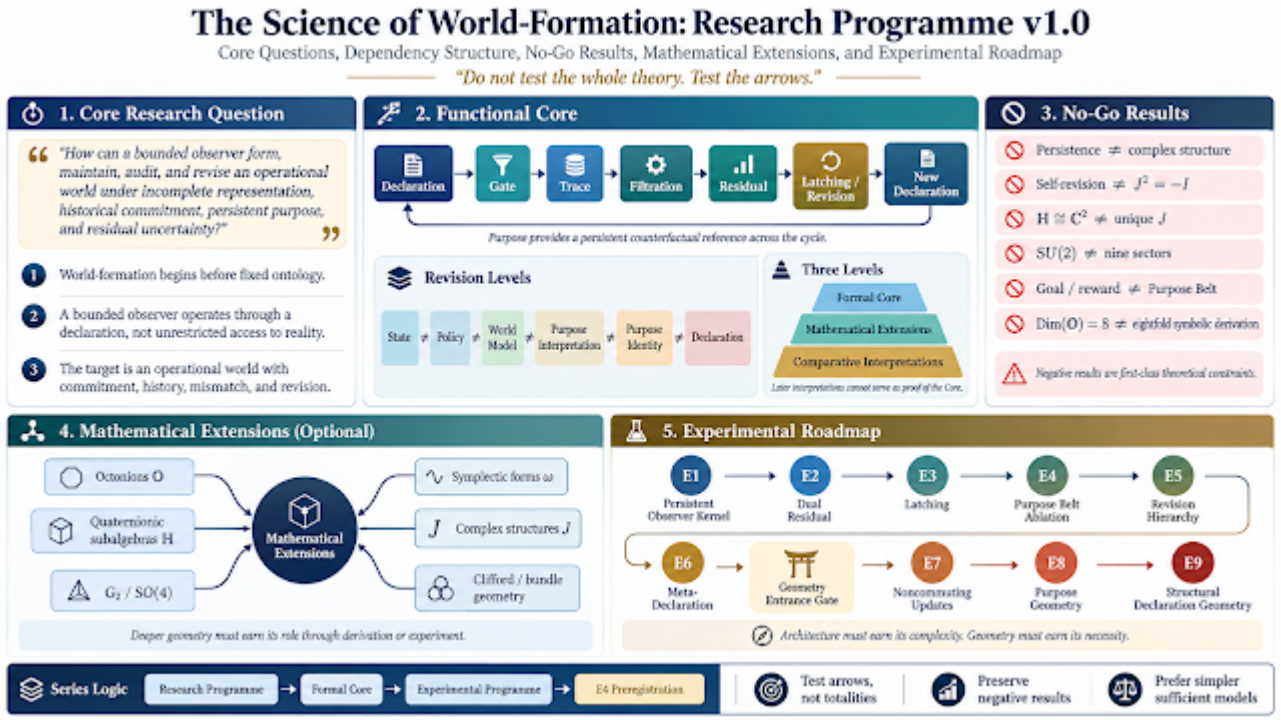
Three levels are kept strictly separate. The **Formal Core** contains the minimal functional architecture. **Mathematical Extensions** include candidate structures such as octonionic carriers, quaternionic subalgebras, $G_2/SO(4)$ declaration spaces, symplectic forms, compatible complex structures, Clifford constructions, and bundle geometry. **Comparative Interpretations** may later compare independently derived structures with historical or philosophical systems, but such comparisons cannot serve as proofs of the Core. This separation is explicit in the source development of the programme. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

A defining methodological feature is the preservation of negative results. Persistence alone does not imply complex structure. Self-revision alone does not imply $J^2 = -I$. The real-vector-space equivalence $\mathbb{H} \cong \mathbb{C}^2$ does not select a unique complex structure. SU(2) does not determine a nine-sector coarse graining. A goal or reward does not by itself constitute a persistent Purpose architecture. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

The programme therefore proceeds by testing individual dependency arrows rather than demanding acceptance of a total theory. Its central methodological rule is:

> **Do not test the whole theory. Test the arrows.**

The research programme is successful only to the extent that its proposed structures prove formally necessary, experimentally useful, behaviourally irreducible, or predictively productive.



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1. Introduction

1.1 The problem of world-formation

Many theories begin with a world already given.

A state space is specified. Variables are defined. Dynamics act on those variables. Observers are introduced later as entities that measure, infer, control, or interpret what already exists.

The Science of World-Formation begins one step earlier.

It asks:

> **What must a bounded system possess before there is, for that system, a stable operational world within which observation, action, memory, error, and revision can meaningfully occur?**

This is not the claim that external reality depends on an observer.

The narrower claim is methodological:

A bounded observer never operates directly on unrestricted possibility. It operates through some finite declaration of what counts as relevant state, admissible distinction, legitimate evidence,

possible action, and meaningful historical consequence.

Accordingly, an operational world is not merely a collection of states.

It is a governed closure.

A first working definition is therefore:

> **An operational world is a structured domain in which distinctions, transitions, commitments, records, residuals, and revisions can be jointly maintained by a bounded observer.**

This shifts attention from ontology alone to the architecture by which an observer acquires and preserves a world.

1.2 The foundational question

The central question of the programme is:

> **What structures are required for a bounded system not merely to operate inside a world, but to form, maintain, audit, and revise an operational world of its own?**

The corresponding research problem can be written schematically as:

Possibility → Declaration → Operational World → Trace → History → Residual → Revision. (1.1)

This sequence is not assumed to be the only possible formulation.

It is the initial dependency skeleton to be formalised, challenged, reduced, and tested.

1.3 What this programme is not

The programme does not begin by asserting that the world is fundamentally:

octonionic;

quaternionic;

complex;

symplectic;

gauge-theoretic;

computational;

informational;

semantic;

or governed by any particular traditional symbolic system.

Those may become useful extensions.

They are not the starting assumptions.

The source development explicitly separates the functional Core from mathematical extensions such as Octonions, G2/SO(4), quaternionic subalgebras, symplectic and complex geometry, Clifford structures, bundles, connections, and holonomy. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

The programme therefore adopts a strong asymmetry:

> **A deeper mathematical structure may explain a validated functional architecture, but it may not be used retrospectively to justify that architecture merely because the correspondence is elegant.**

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2. Research Contract

A research programme of this scope is especially vulnerable to attractive over-unification.

The first task is therefore not to maximise explanatory reach.

It is to impose constraints.

2.1 Epistemic-status ledger

Every major statement should belong to one of the following classes.

Primitive
A role treated as irreducible within the current Core.

Assumption
A condition explicitly introduced rather than derived.

Known Mathematics
A mathematical fact imported from established theory.

Derived Result
A conclusion proved from stated assumptions.

Construction
A useful realisation that is not claimed to be unique or necessary.

Hypothesis
A claim awaiting formal or empirical testing.

No-Go Result
A demonstrated failure of a weaker assumption to imply a stronger structure.

Superseded Claim
An earlier formulation replaced or downgraded by later analysis.

Interpretation
A philosophical, historical, or cross-domain comparison that does not establish the Core.

The source development explicitly calls for this ledger-based discipline and for rebuilding the dependency graph accordingly. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

2.2 Analogy is not derivation

Structural resemblance is not proof.

Equal dimension is not proof.

Equal cardinality is not proof.

Shared notation is not proof.

For example:

$\dim_{\mathbb{R}}(\mathbb{O}) = 8.$ (2.1)

This fact alone cannot derive an eightfold symbolic classification.

Likewise:

$\mathbb{H} \cong \mathbb{C}^2$ as real vector spaces. (2.2)

This does not select a unique complex structure J.

The programme therefore distinguishes:

structural compatibility ≠ mathematical necessity. (2.3)

A correspondence becomes theoretically significant only when a non-arbitrary, structure-preserving map or independent selection principle can be established.

2.3 Negative results are first-class results

A mature theory must preserve the arrows that fail.

The current programme already contains several important No-Go results:

Persistence ⇏ complex structure. (2.4)

Self-revision ⇏ $J^2 = -I$. (2.5)

$\mathbb{H} \cong \mathbb{C}^2$ ⇏ unique J. (2.6)

SU(2) ⇏ nine sectors. (2.7)

Goal or reward ⇏ Purpose Belt. (2.8)

These are not embarrassments.

They constrain future derivation.

The source material explicitly identifies such results as signs of theoretical maturation rather than failures to be hidden. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

2.4 Simpler successful models defeat stronger claims

If a proposed component can be removed without changing the relevant action or revision behaviour, that component has not demonstrated functional irreducibility.

Formally, let B denote a proposed architecture and Z a lower-complexity representation.

If, for every relevant history h,

$P(A_{t:t+k}, U_{t:t+k} \mid Z, h) = P(A_{t:t+k}, U_{t:t+k} \mid B, h)$, (2.9)

then the additional structure in B is behaviourally redundant with respect to the tested domain.

The source material proposes this directly as a minimality criterion for the Purpose architecture: if a smaller state representation preserves both actions and revision decisions, the removed component is bookkeeping rather than behaviourally irreducible information. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

This principle generalises beyond Purpose.

Every architectural component must earn its complexity.

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3. The Minimal Domain

3.1 Possibility

Let Σ denote a domain of possibilities not yet fully organised under a specific operational declaration.

No physical ontology is implied.

Σ may represent:

candidate states;

possible observations;

semantic alternatives;

hypotheses;

environmental configurations;

model classes;

or a more abstract possibility carrier.

The only initial requirement is that a bounded observer does not have unrestricted operational access to all of Σ.

3.2 Bounded observer

A bounded observer O is a system with finite access, finite representation, finite memory, finite action capability, or finite revision capability.

Schematically:

$O(\Sigma) \neq \Sigma.$ (3.1)

Boundedness creates the need for selective representation.

That need is the starting point for Declaration.

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4. Declaration

4.1 Definition

A Declaration D specifies which distinctions, relations, operations, and evidence are admitted into the observer's current operational world.

Write:

$D: \Sigma \rightarrow W_D.$ (4.1)

Here W_D is the world admitted under Declaration D.

Declaration should not be understood merely as verbal assertion.

It may include:

representation choice;

state-variable selection;

ontology;

admissible action class;

evidence policy;

model class;

boundary conditions;

observational interface;

revision constraints.

Thus:

Declaration = operational world selection. (4.2)

The source development explicitly elevates Declaration as model/world selection and distinguishes ordinary learning inside D_n from a higher-order transition $D_n \to D_{n+1}$ in which the representation or model class itself changes. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

4.2 Ordinary learning versus declaration revision

Suppose θ denotes ordinary parameters within a current model.

Ordinary learning has the form:

$\theta_n \to \theta_{n+1}$ while D remains fixed. (4.3)

Declaration revision instead has the form:

$D_n \to D_{n+1}$. (4.4)

The distinction is essential.

Changing parameters inside a representation is not the same as changing the representation through which the world is made operational.

This motivates a hierarchy:

State update → Policy update → Purpose update → Declaration update. (4.5)

The source explicitly proposes these as distinct revision classes with distinct gates and costs. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

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5. Gate, Trace, and Filtration

5.1 Gate

A bounded observer may encounter many candidate observations without committing all of them as historically binding facts.

A Gate G determines whether a candidate outcome becomes part of the observer's operative history.

Write:

$g_n = G(z_n \mid D_n, P_n, F_n)$. (5.1)

In the simplest binary case:

$g_n \in \{0, 1\}$. (5.2)

The conceptual role of Gate is to distinguish:

possible observation ≠ committed historical outcome. (5.3)

5.2 Trace

A committed outcome becomes a Trace T_n.

A simple construction is:

$T_n = g_n \odot z_n$. (5.4)

The exact algebra is implementation-dependent.

The important point is functional:

Trace is observation after commitment.

A Trace can influence future action, future interpretation, and future revision.

5.3 Filtration

Committed traces accumulate.

Let F_n denote the observer's available historical ledger at step n.

Then:

$F_0 \subseteq F_1 \subseteq F_2 \subseteq \dots$ (5.5)

This filtration structure represents increasing historical disclosure.

The source development repeatedly treats Trace and Filtration as central components of the observer architecture and later reuses them in the reduced Purpose architecture rather than duplicating a second complete memory system. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1...

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6. Residual

6.1 Definition

No finite declaration is assumed to absorb every relevant feature of encountered reality or evidence.

Residual R records what the current declaration cannot adequately reconcile.

Abstractly:

$R_n = \text{Residual}(D_n, P_n, F_n)$. (6.1)

Residual should not be treated simply as generic error.

A useful distinction is between magnitude and direction.

6.2 Residual magnitude

Let:

$\mathscr{R}_n \geq 0 \quad (6.2)$

represent the total degree of mismatch.

A simple construction is:

$\mathscr{R}_n = \Sigma_{k \leq n} w_k \|e_k\|^2. \quad (6.3)$

This answers:

How much mismatch has accumulated?

6.3 Directional residual

A second quantity may capture persistent directional structure in error.

Write:

$\mathcal{G}_n = \Sigma_{k \leq n} a_k e_k. \quad (6.4)$

or, on a declaration manifold $\mathscr{M}$_D:

F_R ∈ T_D$\mathscr{M}$_D. (6.5)

This answers a different question:

Is the mismatch systematically pointing toward a particular structural revision?

The distinction matters because:

large residual magnitude + near-zero direction may indicate noise, (6.6)

while:

small repeated errors + persistent direction may indicate structural bias. (6.7)

The source explicitly develops this dual-residual distinction and argues that it may be particularly important for self-revising agents. $\mathbb{O} \to G_2_SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1...

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7. Latching

7.1 Historical resistance

If every observation could costlessly rewrite the observer's Purpose, policy, or world model, the system would possess maximal plasticity but almost no historical identity.

Latching introduces revision resistance.

Let κ_ℓ denote the switching cost associated with revision level ℓ.

A generic revision rule is:

revise at level ℓ only if $\Delta L_\ell > \kappa_\ell$. (7.1)

Otherwise:

Latch. (7.2)

This creates:

plasticity + persistence. (7.3)

The source proposes precisely this finite regime: Purpose should be revisable, but not casually revisable. $\mathbb{O} \to G_2_SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1…

7.2 Hierarchical latching

A natural working hypothesis is:

$\kappa_{policy} < \kappa_{world} < \kappa_{interpretation} < \kappa_{Purpose}$. (7.4)

This is not yet a universal law.

It is an experimentally testable architecture.

The intended interpretation is that rewriting a high-level persistent commitment should normally require stronger evidence than changing a low-level action rule.

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8. Purpose

8.1 Goal is not Purpose

A Goal may be represented by a scalar objective:

$\min_x L(x)$. (8.1)

A system may optimise such an objective without preserving a distinction between:

what it was originally trying to achieve;

how that objective is currently interpreted;

what actually occurred;

why the interpretation changed;

whether the revised interpretation remains a legitimate continuation of the original commitment.

The programme therefore distinguishes Goal from Purpose.

Goal ≠ Purpose. (8.2)

The source explicitly rejects the identification:

Purpose = scalar reward, (8.3)

and also rejects reducing Purpose to a mere system prompt. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

8.2 Persistent counterfactual reference

The minimal functional role of Purpose is to provide a persistent reference that remains distinguishable from realised history.

Write:

P_ref ≠ H_realised. (8.4)

This allows the system to ask not only:

> What happened?

but also:

> Does what happened remain compatible with what the system is still trying to make possible?

This separation is central to the Purpose Belt hypothesis.

The source later sharpens the claim into a requirement for persistent, auditable separation between Purpose identity, current interpretation, realised history, and the rules governing revision. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

8.3 Purpose identity and Purpose interpretation

A major refinement is the distinction:

Purpose Identity ≠ Current Purpose Interpretation. (8.5)

Let:

P_n = persistent Purpose identity, (8.6)

and:

I_n = interpretation of P_n under the current world model W_n. (8.7)

Then an ontology shift may require:

$I_n \rightarrow I_{n+1}$, (8.8)

while preserving:

$P_{n+1} \approx P_n$. (8.9)

This allows a system to adapt its interpretation without silently replacing the underlying Purpose.

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9. Revision Attribution

Residual alone does not determine what should change.

Suppose discrepancy r is detected.

The agent may need to distinguish among at least:

policy failure, (9.1)

world-model failure, (9.2)

Purpose-interpretation failure, (9.3)

Purpose-identity failure. (9.4)

Let:

$\ell \in \{\pi, W, I, P\}$. (9.5)

Revision Attribution estimates:

$A(r) = P(\ell \mid r, F, D, P)$. (9.6)

Then:

$\ell = \pi \Rightarrow$ revise policy, (9.7)

$\ell = W \Rightarrow$ revise world model, (9.8)

$\ell = I \Rightarrow$ revise Purpose interpretation, (9.9)

$\ell = P \Rightarrow$ revise Purpose identity. (9.10)

The source identifies this as one of the most important functional additions: Purpose Belt becomes nontrivial only when different diagnoses lead to genuinely different revision classes. $\mathbb{O} \to G_2 \to SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1...

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10. The Minimal Purpose–Observer Kernel

The programme does not require a large literal "belt."

Later analysis compresses the architecture substantially.

The Self-Referential Observer already provides:

Trace + Filtration + Adaptive Policy + Latching. (10.1)

The Purpose layer need only add:

Purpose Identity + Interpretation + Revision Attribution + Purpose-level Latching. (10.2)

This leads to a leaner architecture:

Self-Referential Observer + Purpose Interpretation Layer + Revision Governor. (10.3)

The source explicitly proposes this reduced form rather than treating the observer and Purpose Belt as two independent large subsystems. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

A minimal state candidate is therefore:

$B_n = (P_n, I_n, W_n, H_n, A_n; \kappa)$. (10.4)

where:

P_n = Purpose identity,
I_n = Purpose interpretation,
W_n = world model,
H_n = sufficient historical state,
A_n = revision-attribution state,
κ = revision-cost structure.

The source also notes that full historical trajectories need not be stored explicitly if sufficient statistics preserve relevant action and revision behaviour. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

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11. Canonical World-Formation Cycle

The programme can now be summarised as a recursive operational cycle.

Let the world-forming state be:

$\Omega_n = (D_n, P_n, x_n, F_n, R_n)$. (11.1)

Operational dynamics produce a new state:

$x_{n+1} = \Phi_D_n,P_n(x_n, u_n, \xi_n)$. (11.2)

Observation produces a candidate:

$z_n = H_D_n(x_n)$. (11.3)

Gate determines commitment:

$g_n = G(z_n \mid D_n, P_n, F_n)$. (11.4)

Committed observation becomes Trace:

$T_n = g_n \odot z_n$. (11.5)

Trace updates Filtration:

$F_{n+1} = F_n \vee T_n$. (11.6)

Residual is evaluated:

$R_{n+1} = E(D_n, P_n, F_{n+1})$. (11.7)

Revision level is attributed:

$\ell_{n+1} = A(R_{n+1}, D_n, P_n, F_{n+1})$. (11.8)

Revision then acts:

$(D_{n+1}, P_{n+1}) = U^{(\ell_{n+1})}(D_n, P_n, F_{n+1}, R_{n+1})$. (11.9)

The conceptual cycle is therefore:

Declaration → Dynamics → Gate → Trace → Filtration → Residual → Attribution → Latching / Revision → New Declaration. (11.10)

The defining recursive feature is:

> **The history produced under a declaration can later participate in revising the declaration through which that history became meaningful.**

This is the programme's minimal form of recursive world-formation.

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12. Assumption Dependency Structure

The source development proposes an explicit dependency graph rather than a single uninterrupted narrative. $\mathbb{O} \to G_2_SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1…

A working version is:

A1. Finite persistent boundary → Gate / memory pressure. (12.1)

A2. Imperfect representation → Residual. (12.2)

A3. Nonzero revision cost → Latching. (12.3)

A4. Persistent counterfactual Purpose → Reference / realisation duality. (12.4)

A5. Accountable orientation → candidate antisymmetric relation. (12.5)

A6. Nondegeneracy → candidate symplectic structure. (12.6)

A7. Positive Purpose metric → candidate compatible complex structure. (12.7)

A8. Real dimension four + compatible J → complex dimension two. (12.8)

A9. Quaternionic compatibility → restricted admissible J-family. (12.9)

The graph is deliberately heterogeneous.

The early assumptions belong near the functional Core.

The later assumptions belong to mathematical extension.

This separation prevents one attractive mathematical structure from silently carrying assumptions that should have been independently justified.

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13. The No-Go Ledger

The programme formally records several conclusions that **cannot** presently be derived from weaker assumptions.

NG0 — Real self-revision is possible

A Gate–Memory–Residual–Self-Revision system can operate entirely over real scalar dynamics.

Therefore:

Adaptive self-revision $\not\Rightarrow$ complex observer structure. (13.1)

The source explicitly treats this as a valuable negative result. $\mathbb{O} \to G_2_SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1…

NG1 — Persistence is insufficient

Persistence $\not\Rightarrow$ complex structure. (13.2)

NG2 — Self-revision is insufficient

Self-revision $\not\Rightarrow J^2 = -I$. (13.3)

NG3 — $\mathbb{H} \cong \mathbb{C}^2$ does not determine J

$\mathbb{H} \cong \mathbb{C}^2 \not\Rightarrow$ unique J. (13.4)

NG4 — Orthogonal J is not automatically quaternionically admissible

$J^2 = -I$ and metric compatibility do not by themselves establish the required quaternionic polarisation.

NG5 — Circular phase does not force four-state coarse graining

$S^1 \not\Rightarrow$ four-state partition. (13.5)

NG6 — SU(2) does not force nine sectors

$SU(2) \not\Rightarrow N = 9$. (13.6)

NG7 — Octonionic dimension does not derive an eightfold symbolic system

$\dim_{\mathbb{R}}(\mathbb{O}) = 8 \not\Rightarrow$ eight-category interpretive structure. (13.7)

NG8 — Goal does not imply Purpose Belt

Goal / Reward $\not\Rightarrow$ persistent Purpose architecture. (13.8)

These constraints define what later papers must not claim without new derivation or evidence.

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14. Programme Thesis

The minimal thesis of World-Formation Theory is intentionally modest.

It is not:

> Reality is fundamentally described by a particular algebra.

It is:

> **A bounded observer requires more than a state space if it is to possess an operational world with commitment, history, mismatch, persistence, and self-revision.**

The proposed minimal loop is:

Declaration → Gate → Trace → Filtration → Residual → Latching → Revision → Declaration. (14.1)

Purpose adds a further structure:

Persistent counterfactual reference ≠ realised history. (14.2)

Together, these structures create the possibility of a system that can ask not only:

> What state am I in?

but also:

> What world am I currently using?
>
> What has been committed into its history?
>
> What no longer fits?
>
> What level should be revised?

> Which commitments should survive the revision?

That is the initial scientific content of the programme.

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15. Transition to the Formal Core

The next document in the series, **World-Formation Formal Core v1.0**, removes most programme-level discussion and develops the minimal architecture in a stricter form.

Its task is not to make the theory larger.

Its task is to make the theory harder to evade.

The Formal Core will therefore attempt to determine:

which objects are primitive;
which relations are assumptions;
which components are behaviourally irreducible;
which claims are merely constructions;
which arrows remain conjectural;
and which stronger structures have already been ruled out by No-Go results.

The guiding principle remains:

> **Architecture must earn its complexity.**

And, for all later mathematical development:

> **Geometry must earn its necessity.**

Continue the research programme

Write the Experimental Programme v1.0

Write the Formal Core v1.0

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive

scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.